Exam Date & Time: 21-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech. V Semester Automobile Engineering- End Semester Examination - Nov-Dec 2024

HEAT TRANSFER [AAE 3128]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

* Answer all questions.
* Use of Heat and Mass Transfer Data Hand Book is allowed
* Assume the missing data suitably.
* Write neatly and legibly.

1) A 4-mm-diameter spherical ball at 50°C is covered by a 1 mm thick plastic insulation (k = 0.15 W/m/K). The ball is exposed to a medium at 15° C, with a combined convection and radiation heat transfer coefficient of 20 W/m2/K. Analyze whether the insulation is effective or not. (3)

2) A 4-mm-diameter and 10-cm-long aluminum fin (k = 220 W/m/K) is attached to a surface. If the heat transfer coefficient is 10 $W/m^2/K$, determine the percent error in the rate of heat transfer from the fin when the infinitely long fin assumption is used instead of the adiabatic fin tip assumption. (4)

3) Two solid cylinders of identical lengths are made of Tungsten and Tin respectively. Assuming axial heat conduction across the cylinders, find out the ratio of their diameters when the thermal resistances of conduction for both the cylinders are equal. (3)

4) Hot exhaust gases of a stationary diesel engine are to be used to generate steam in an evaporator. Exhaust gases (c_p = 1051 J/kg/K) enter the heat exchanger at 540°C at a rate of 0.25 kg/s while water enters as saturated liquid and evaporates at 200°C (h_{fg} = 1941 kJ/kg). The heat transfer surface area of the heat exchanger based on the water side is 0.5 m^2, and the overall heat transfer coefficient is 1780 $W/m^2/K$. Determine the rate of heat transfer, the exit temperature of the exhaust gases, and the rate of evaporation of the water. (4)

5) Consider a long rectangular fin attached to a flat surface, with its end temperature nearly equal to the surrounding air temperature (20°C). The fin has a width of 5.0 cm, a thickness of 1.0 mm, a thermal conductivity of 200 W/m/K, and a base temperature of 40°C. The heat transfer coefficient is 20 $W/m^2/K$. Estimate the temperature of the fin at a point 5 cm from the base, as well as the total rate of heat loss from the fin. (4)

6) Describe the following dimensionless numbers and discuss their significance: (i) Peclet Number (ii) Stanton Number. (2)

7) A plate measuring 1 m in length and 0.5 m in width is positioned lengthwise in a stream of crude oil flowing at 25 m/s. Given that the oil has a specific gravity of 0.72 and a kinematic viscosity of 1.1 stokes, calculate the following:

i. The boundary layer thickness at the midpoint of the plate

ii. The shear stress at the center of the plate

iii. The frictional drag on one side of the plate. (4)

8) Flue gases from an incinerator are released to atmosphere using a stack that is 0.6 m in diameter and 10.0 m high. The outer surface of the stack is at 40° C and the surrounding air is at 10° C. Determine the rate of heat transfer from the stack. (3)

9) Analyze the salient characteristics of nucleate boiling. (3)

10) What is fouling factor in heat exchangers? (2)

11) A 10-meter-long copper tube with a 25 mm diameter is used to transport cold coolant, which keeps the tube surface temperature at 5°C. This temperature difference causes moisture in the surrounding air to condense on the tube's outer surface. Calculate the rate of dropwise condensation of the moisture from the air, given a saturation temperature of 25°C. (3)

12) Derive an expression for the heat transfer between two infinitely large grey plates. (5)

13) Consider a person whose exposed surface area is 2 m^2, emissivity is 0.85, and surface temperature is 32°C. Determine the rate of heat loss from that person by radiation in a large room whose walls are at a temperature of 25°C. (3)

14) Analyze the behaviour of the finely sprayed fuel mist at the end of the compression stroke in a compression ignition engine. (4)

15) Consider two black coaxial parallel circular disks of equal diameter D that are spaced apart by a distance L. The top and bottom disks have uniform temperatures of 500° C and 520° C, respectively. Determine the radiation heat transfer between the disks if they are spaced apart by L = D. (3)

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